

SUMMARY

- **Profile:** Currently pursuing a Bachelor’s in Artificial Intelligence and Data Science, with a strong interest in machine learning, data analysis, and software development. Skilled in Python, Java, and data visualization, and eager to learn and apply AI and programming concepts to solve real-world problems. Passionate about building innovative, data-driven solutions and growing as a software professional.

EDUCATION

Degree	Institution	Year of Passing	CGPA/Marks
B.Tech in Artificial Intelligence and Data Science	Amrita Vishwa Vidyapeetham, Coimbatore	2027	6.89 / 10
Intermediate (Class XII)	Sri Chaitanya Junior College , vijayawada	2023	92%
Secondary School (Class X)	Z.P.H. School	2021	93%

RESEARCH PUBLICATIONS

- **Identification and Sorting of Objects by Color using Robotic Arm:** Patnayakuni Hari Gopal Vatsal, Vattikuti Sri Mahesh Abhiram, S Janaki Ramamjaneya Reddy, Mohit Madhav Kuthadi. *2025 6th International Conference on Recent Advances in Information Technology (RAIT)*, March 2025. DOI: 10.1109/RAIT65068.2025.11088895

PROJECTS

- Color Sorting Robotic Arm

Raspberry Pi, CNN, OpenCV

 - Implemented Raspberry Pi-based vision pipeline for object color detection using CNN / color segmentation.
 - Controlled a robotic arm to automatically pick and place objects into color-based bins for sorting.
- Plant Disease Detection using Big Data Tools

Hadoop, Spark, Python

 - Built a scalable distributed plant disease detection system using Hadoop and Apache Spark.
 - Leveraged MobileNetV2 for lightweight CNN-based image classification with a distributed preprocessing and inference pipeline for large-scale agricultural datasets.
- Plant Disease Detection Vehicle

Raspberry Pi/Jetson, GPS, CNN

 - Developed an autonomous rover that captures crop images and detects plant diseases in real time.
 - Integrated GPS to log affected regions and provided treatment suggestions for precision agriculture.
- Smart Blind Hat

Computer Vision, PyTesseract, DeepSORT

 - Developed a wearable device with a head-mounted camera to detect and track moving objects.
 - Provided real-time audio guidance for navigation and text reading to assist visually impaired users.
- Hand Gesture Controlled Robotic Arm

Arduino, Flex Sensors, Accelerometer

 - Controlled a multi-DOF robotic arm using hand gestures captured via flex and accelerometer sensors.
 - Implemented smooth servo control for real-time mapping of human hand motion to robot movement.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL

Web Technologies: HTML, CSS

AI & Machine Learning: Machine Learning, Deep Learning, Neural Networks, Model Training & Evaluation

Advanced AI Models: Large Language Models (LLMs)

Frameworks & Libraries: NumPy, TensorFlow, PyTorch

Big Data & Tools: Hadoop, Git, GitHub

Databases: SQL